Universal Recovery: The Petz Map as Retrodiction Functor, Error Corrector, and Entropy Bound

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The Petz recovery map appears independently in quantum sufficiency theory, quantum error correction, categorical retrodiction, and quantum thermodynamics. We prove that these appearances are not analogies but manifestations of a single mathematical structure. Starting from the quantum eraser—where "erased" which-path information is recovered via post-selection—we show that this recovery is precisely the Petz map, which is the *unique* retrodiction functor satisfying Bayesian consistency (Theorem 1). This uniqueness implies near-optimal quantum error correction when approximate Knill–Laflamme conditions hold (Theorem 2). When the reference state is thermal, the Gibbs structure $\sigma^{1/2} = e^{-\beta H/2}/\sqrt{Z}$ yields a chain of inequalities binding recovery fidelity to entropy production (Theorem 3): $-\log F^2 \leq I(A;E|B) \leq \Sigma \leq \Delta D$. We define a temporal asymmetry parameter $\tau = 1 - F$ that quantifies the degree to which a quantum process distinguishes past from future: $\tau = 0$ characterizes closed, unitary evolution where forward and reverse processes are statistically indistinguishable; $\tau > 0$ measures openness to an environment. The quantum eraser, Bayesian retrodiction, error correction, and the thermodynamic arrow of time are four aspects of one mathematical object—the Petz recovery map—and the arrow of time is the statement $\tau > 0$.

INTRODUCTION

The quantum eraser experiment [1, 2] reveals a remarkable fact: which-path information that appears to have been irreversibly acquired can be "erased" by appropriate measurement on an entangled partner, restoring interference. Operationally, given a quantum channel $\mathcal{N}$ (the which-path marking), the question is whether a recovery map $\mathcal{R}$ exists such that $\mathcal{R} \circ \mathcal{N} \approx \mathrm{id}$ on a relevant subspace. This is simultaneously the question of the quantum eraser, quantum error correction (QEC), and Bayesian retrodiction.

The Petz recovery map [3, 4]—originally introduced as a technical tool for sufficiency of quantum subalgebras—has been independently rediscovered in each of these contexts: as a near-optimal QEC decoder [5]; as the *unique* retrodiction functor in categorical quantum mechanics [6]; as the reverse channel in quantum fluctuation theorems [7]; as the physical reversal of irreversible processes via dilations [17]; and as the backbone of entanglement wedge reconstruction in holography [8]. These communities have largely not recognized that the same algebraic object governs all four phenomena.

In this Letter, we establish this unification rigorously. We assemble three known but previously disconnected results into a single framework: (1) the quantum eraser is a physical instantiation of the Petz recovery, whose categorical uniqueness as a retrodiction functor (Theorem 1, due to Parzygnat–Buscemi [6]) forces it to be the canonical "time reversal" in quantum mechanics; (2) this uniqueness implies near-optimal QEC when approximate Knill–Laflamme conditions hold (Theorem 2, combining [5, 11]); and (3) the Gibbs-state structure of the reference state embeds the second law of thermodynamics as a recoverability constraint, yielding a master inequality chain (Theorem 3, assembling [11–13, 18]). The conceptual contribution is the identification that these four phenomena share one mathematical object, and the construction of the equivalence chain (11) connecting them. The logical flow is:

$$\text{Eraser} \xrightarrow{\text{Petz}} \text{Retrodiction} \xrightarrow{\text{Thm. 1}} \text{QEC} \xrightarrow{\text{Thm. 2}} \text{Thermo.} \tag{1}$$

Beyond this mathematical unification, the framework yields a physical insight: the thermodynamic arrow of time—the empirical asymmetry between past and future—emerges as the failure of Petz recovery. In a closed system undergoing unitary evolution, the Petz map recovers the initial state exactly ($F = 1$), the Crooks ratio $P_F/P_R = \exp(\Sigma/k_\mathrm{B})$ equals unity [26], and forward and reverse processes are statistically indistinguishable. Openness to an uncontrolled environment ($\Sigma > 0$) breaks this symmetry. We quantify this via a temporal asymmetry parameter $\tau = 1 - F(\rho, \tilde{\mathcal{R}}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho)))$ and show that $\tau \leq 1 - \exp(-\Sigma/2)$, vanishing if and only if the process produces zero entropy.

THE PETZ RECOVERY MAP

Given a quantum channel $\mathcal{N} : \mathcal{B}(\mathcal{H}_A) \to \mathcal{B}(\mathcal{H}_B)$ and a faithful reference state $\sigma \in \mathcal{S}(\mathcal{H}_A)$, the *Petz recovery map* is

$$\mathcal{R}_{\sigma,\mathcal{N}}(\rho) = \sigma^{1/2}\, \mathcal{N}^\dagger\big(\mathcal{N}(\sigma)^{-1/2}\, \rho\, \mathcal{N}(\sigma)^{-1/2}\big)\, \sigma^{1/2}, \tag{2}$$

where $\mathcal{N}^\dagger$ is the Hilbert–Schmidt adjoint of $\mathcal{N}$. The map $\mathcal{R}_{\sigma,\mathcal{N}}$ is completely positive and trace-preserving (CPTP) [3], and satisfies exact recovery $\mathcal{R}_{\sigma,\mathcal{N}} \circ \mathcal{N} = \mathrm{id}$ on the support of σ if and only if the data-processing inequality (DPI) for relative entropy is saturated: $D(\sigma\|\tau) = D(\mathcal{N}(\sigma)\|\mathcal{N}(\tau))$ for all τ [4].

Table I shows how the mathematical ingredients of Eq. (2) map to physical meaning in each domain.

TABLE I. Physical interpretation of the Petz map $\mathcal{R}_{\sigma,\mathcal{N}}$ across four domains.

	Eraser	**Time Rev.**	**QEC**	**Thermo.**
$\mathcal{N}$	Path mark	Fwd. evol.	Noise	Irrev. proc.
σ	Entangled	Prior	Code state	Gibbs
$\mathcal{R}$	Erasure	Retrodiction	Decoder	Rev. proc.
ΔD	Decoherence	Info. loss	Code dist.	βW_{diss}

THEOREM 1: UNIQUENESS OF RETRODICTION

The Petz map is not merely *a* recovery map but *the unique* one satisfying natural axioms of Bayesian reasoning.

Theorem 1 (Retrodiction Uniqueness [6])**.** *Let* C *be the category of finite-dimensional* C^**-algebras with state-preserving CP maps as morphisms. A* retrodiction functor *is a contravariant functor* $R : \mathsf{C} \to \mathsf{C}^{\mathrm{op}}$ *satisfying:*

(i) Preservation: *R sends $(\mathcal{N}, \sigma)$ to a CPTP map.*

(ii) Bayesian consistency: *For composable channels $\mathcal{N}_1, \mathcal{N}_2$ and prior σ,*

$$R(\mathcal{N}_2 \circ \mathcal{N}_1,\, \sigma) = R(\mathcal{N}_1,\, \sigma) \circ R(\mathcal{N}_2,\, \mathcal{N}_1(\sigma)). \tag{3}$$

(iii) Normalization: $R(\mathrm{id}, \sigma) = \mathrm{id}$.

(iv) Classical limit: *On commutative subalgebras, R reduces to classical Bayes' rule.*

Then $R(\mathcal{N}, \sigma) = \mathcal{R}_{\sigma,\mathcal{N}}$ is the unique such functor.

Proof sketch.—Axioms (ii) and (iv) force the $\sigma^{1/2}(\cdot)\sigma^{1/2}$ sandwich structure. For commutative algebras, R must be the Bayesian inverse $R(N, p)(q)_i = p_i N_{j|i}/(\sum_k p_k N_{j|k})$, which in matrix form is $p^{1/2} N^T (Np)^{-1/2}(\cdot)(Np)^{-1/2} N p^{1/2}$. By the functoriality axiom (3) and the Stinespring dilation theorem, this extends uniquely to the noncommutative setting as Eq. (2). The full proof is in Ref. [6]; see also [9] for the Bayesian framework and Supplemental Material [31] for a self-contained derivation. □

Quantum eraser as Petz recovery.—In the quantum eraser, a signal-idler pair is prepared in the entangled state $|\psi\rangle_{SE} = (|A, d_A\rangle + |B, d_B\rangle)/\sqrt{2}$, where $|A\rangle, |B\rangle$ are path states and $|d_A\rangle, |d_B\rangle$ are orthogonal detector states. The which-path channel is $\mathcal{N} = \mathrm{Tr}_E$, giving the fully decohered signal state $\rho_S = \frac{1}{2}(|A\rangle\langle A| + |B\rangle\langle B|)$. Applying the Petz recovery (2) with $\sigma = |\psi\rangle\langle\psi|_{SE}$ and $\mathcal{N} = \mathrm{Tr}_E$:

$$\mathcal{R}_{\sigma,\mathrm{Tr}_E}(\rho_S) = |\psi\rangle\langle\psi|_{SE}, \tag{4}$$

which is exact because Tr_E saturates the DPI for pure bipartite states [31]. The "erasure measurement" on the idler (projecting onto $(|d_A\rangle \pm |d_B\rangle)/\sqrt{2}$) is the physical implementation of this Petz recovery. Strictly, the idler measurement is a *quantum instrument* (a collection of CP maps summing to a CPTP map); conditioned on outcome $\pm$, the effective map on the signal is CP but not trace-preserving. The Petz map $\mathcal{R}_{\sigma,\mathrm{Tr}_E}$ itself is CPTP and recovers the full entangled state (4); the subsequent projection onto $|\pm\rangle$ selects the subensemble where the coherent superposition $|A\rangle \pm |B\rangle$ is restored. This identification has not been stated as a formal theorem in prior literature. The functoriality axiom (3) corresponds to the sequential structure of the eraser: composing the which-path marking with the idler measurement decomposes as retrodiction from the idler, then from the path.

Connection to TSVF.—The Aharonov–Bergmann–Lebowitz two-state vector formalism [14] assigns both a forward-evolving state $|\psi\rangle$ and a backward-evolving state $\langle\phi|$. Theorem 1 provides the unique canonical choice for the backward state: it is the Petz-retrodicted state. This resolves the ambiguity in TSVF about which backward state to use.

THEOREM 2: FROM RETRODICTION TO ERROR CORRECTION

The uniqueness of the Petz map as retrodiction functor *implies* its near-optimality as a QEC decoder.

Theorem 2 (Retrodiction Implies Near-Optimal QEC)**.** *Let $\mathcal{N}$ be a noise channel acting on a $[\![n, k, d]\!]$ stabilizer code $\mathcal{C}$ with code projector P and maximally mixed code state $\sigma_{\mathcal{C}} = P/\dim\mathcal{C}$. Let $\epsilon_{\mathrm{KL}} = \max_{a,b} \left\| P E_a^\dagger E_b P - c_{ab} P \right\|$ measure deviation from the Knill–Laflamme conditions [10], where $\{E_a\}$ are Kraus operators of $\mathcal{N}$. Then:*

(a) *If $\epsilon_{\mathrm{KL}} = 0$ (exact QEC), $\mathcal{R}_{\sigma_{\mathcal{C}},\mathcal{N}} \circ \mathcal{N} = \mathrm{id}$ on $\mathcal{C}$.*

(b) *In general, by Junge et al. [11], there exists a* rotated *Petz map $\tilde{\mathcal{R}}_{\sigma,\mathcal{N}}$ (a universal recovery map depending only on σ and $\mathcal{N}$) such that:*

$$F\big(\rho,\, \tilde{\mathcal{R}}_{\sigma_{\mathcal{C}},\mathcal{N}}(\mathcal{N}(\rho))\big)^2 \geq \exp\big(-\Delta D\big), \tag{5}$$

where $\Delta D = D(\rho\|\sigma_{\mathcal{C}}) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma_{\mathcal{C}}))$. The standard (non-rotated) Petz map satisfies the weaker Barnum–Knill bound [5]: $F_e \geq \frac{1}{2} F_{e,\mathrm{opt}}^2$.

Proof.—(a) When the Knill–Laflamme conditions hold exactly, the DPI is saturated for all code states: $D(\rho\|\sigma_{\mathcal{C}}) = D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma_{\mathcal{C}}))$ for all ρ supported on $\mathcal{C}$. By Petz's sufficiency theorem [4], the Petz recovery is exact. (b) The rotated Petz map is $\tilde{\mathcal{R}}(\cdot) = \int dt\ p(t)\, \sigma^{it} \mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\sigma)^{-it}(\cdot)\mathcal{N}(\sigma)^{it}) \sigma^{-it}$, where $p(t) =$

$(\pi/2)(\cosh(\pi t)+1)^{-1}$; the bound follows from Ref. [11]. □

The conceptual point is that QEC decoding *is* retrodiction: given the corrupted state $\mathcal{N}(\rho)$, the decoder must "retrodict" the original ρ using the prior $\sigma_{\mathcal{C}}$. By Theorem 1, the Petz map is the *unique* Bayesian-consistent way to do this. Its near-optimality [5] is not a coincidence of linear algebra but a consequence of Bayesian consistency: among all CPTP maps satisfying the functoriality axiom (3), the Petz map is the unique choice, and it achieves fidelity within a universal factor of the optimum.

THEOREM 3: THE THERMODYNAMIC STRUCTURE

When the reference state is thermal, the Petz map encodes the second law of thermodynamics as a constraint on recoverability.

Theorem 3 (Master Inequality Chain)**.** *For any quantum channel $\mathcal{N}$, reference state σ, and state ρ:*

$$\boxed{-\log F^2 \;\le\; I(A;E|B)_\omega \;\le\; \Sigma \;\le\; \Delta D} \tag{6}$$

where $F = F(\rho, \mathcal{R}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho)))$ is the recovery fidelity; $I(A;E|B)_\omega$ is the conditional mutual information of the Stinespring complementary output; Σ is the entropy production (Kolchinsky–Wolpert decomposition [13]); and $\Delta D = D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$ is the relative entropy decrease. When $\sigma = e^{-\beta H}/Z$ is a Gibbs state:

$$\Delta D = \beta\, W_{\rm diss} = \Delta S_{\rm irr}/k_{\rm B}, \tag{7}$$

yielding a thermodynamic bound on recovery: $F^2 \ge e^{-\Delta S_{\rm irr}/k_{\rm B}}$.

Proof.—The chain (6) follows from four established results assembled into a single inequality. Here F denotes the fidelity with the rotated Petz map $\tilde{\mathcal{R}}_{\sigma,\mathcal{N}}$ of Ref. [11].

Step 1 (Fidelity bound): By the universal recovery theorem [11], $F^2 \ge \exp(-\Delta D)$, so $-\log F^2 \le \Delta D$.

Step 2 (CMI bound): By Fawzi–Renner [12], $I(A;E|B) \ge -\log F^2$ via the strengthened data processing inequality applied to the Stinespring dilation, giving the leftmost bound.

Step 3 (Entropy production): For a thermodynamic process with Gibbs reference state $\sigma = e^{-\beta H}/Z$, the entropy production satisfies $\Sigma \ge I(A;E|B)$ because the Landauer component $\Sigma_{\rm Landauer} = D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$ equals $\Delta D \ge I(A;E|B)$ (by Steps 1–2), and $\Sigma \ge \Sigma_{\rm Landauer}$ by the Kolchinsky–Wolpert decomposition [13].

Step 4 (Second law): The relative entropy decrease $\Delta D \ge \Sigma$ by Reeb–Wolf [18]: the free-energy dissipation exceeds the entropy production by system-bath correlations and bath displacement.

The tightness conditions are physically transparent: $F^2 = 1$ iff exact QEC (Knill–Laflamme satisfied); $I(A;E|B) = 0$ iff Markov condition (no information leaks to environment); $\Sigma = 0$ iff thermodynamically reversible process. See Supplemental Material [31] for the detailed derivation including the Gibbs specialization (7). □

Quantum Crooks connection.—When $\sigma = \sigma_\beta = e^{-\beta H}/Z$, the Petz map becomes $\mathcal{R}_{\sigma_\beta,\mathcal{N}}$ with the structural factor $\sigma_\beta^{1/2} = e^{-\beta H/2}/\sqrt{Z}$ —exactly the "half Boltzmann weight" appearing in the KMS condition of modular theory. Kwon and Kim [7] proved that this $\mathcal{R}_{\sigma_\beta,\mathcal{N}}$ is precisely the reverse channel in the quantum Crooks fluctuation theorem:

$$\frac{p_F(w)}{p_R(-w)} = e^{\beta(w-\Delta F)}, \tag{8}$$

where $p_F(w)$ and $p_R(-w)$ are forward and Petz-reversed work distributions. The detailed balance condition $\mathcal{R}_{\sigma,\mathcal{N}} = \mathcal{N}$ characterizes thermodynamically reversible processes—those that are their own Petz reversal.

Temporal asymmetry parameter.—Define

$$\tau \;=\; 1 - F\big(\rho,\, \tilde{\mathcal{R}}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho))\big). \tag{9}$$

This parameter quantifies the degree to which the channel $\mathcal{N}$ distinguishes past from future when σ is the reference state. A remark on the two Petz maps is in order. Theorem 1 singles out the *standard* Petz map $\mathcal{R}_{\sigma,\mathcal{N}}$ as the unique retrodiction functor; the quantum Crooks theorem likewise uses $\mathcal{R}_{\sigma,\mathcal{N}}$ as the physical reverse channel. The bound (5) and the definition of τ require the *rotated* map $\tilde{\mathcal{R}}$, which averages $\mathcal{R}$ over modular automorphisms $\sigma^{it}(\cdot)\sigma^{-it}$. For a thermal reference $\sigma = e^{-\beta H}/Z$, the rotation $\sigma^{it} = e^{-it\beta H}/Z^{it}$ is simply unitary time evolution generated by H, so $\tilde{\mathcal{R}}$ smooths the standard Petz map over the system's own Hamiltonian dynamics. This thermal averaging handles quantum noncommutativity $[\rho,\sigma] \ne 0$ and ensures that the fidelity bound $F^2 \ge \exp(-\Delta D)$ holds universally. When $[\rho,\sigma] = 0$ (classical limit) or $\Delta D = 0$ (exact recovery), $\tilde{\mathcal{R}}$ reduces to $\mathcal{R}$ and the distinction vanishes. Three properties follow from the master chain (6):

(i) *Closed systems:* If $\mathcal{N}$ is unitary, $\mathcal{R}_{\sigma,\mathcal{U}} = \mathcal{U}^\dagger$ and $F = 1$, so $\tau = 0$. There is no temporal asymmetry.

(ii) *Bound by entropy production:* From $F^2 \ge \exp(-\Sigma)$,

$$\tau \;\le\; 1 - \exp(-\Sigma/2). \tag{10}$$

When $\Sigma = 0$ (zero entropy production), $\tau = 0$ is enforced.

(iii) *Equivalence chain:* For faithful σ (see [31], Sec. S11):

$$\tau = 0 \iff \Sigma = 0 \iff I(A;E|B) = 0 \iff \text{QMC}, \tag{11}$$

where QMC denotes the quantum Markov chain condition [12]. Each individual equivalence is established [4,

12, 29]; the full chain (11) states them as a single result, formalizing the physical content: the arrow of time ($\tau > 0$) is equivalent to environmental coupling ($I(A; E|B) > 0$), entropy production ($\Sigma > 0$), and imperfect recoverability ($F < 1$).

PHYSICAL PREDICTIONS

The framework generates two testable predictions that go beyond the mathematical structure established above. While the individual inequalities in Eq. (6) are known results, the following predictions are *new physical consequences* of identifying the Petz map as the unique retrodiction functor (Theorem 1).

Prediction 1: Post-selection as thermodynamic filtering.—For a stabilizer code under noise $\mathcal{N}$, each syndrome s has a data-processing gap $\Delta D(s)$, giving $F(s)^2 \geq \exp(-\Delta D(s))$. Rejecting the fraction ε of syndromes with the largest $\Delta D(s)$ yields $\ln R(\varepsilon, d) = \alpha(p)\, d + \beta(\varepsilon)$, with α (decoder-independent) and β (code-distance-independent). *What is new:* that the per-distance scaling $\alpha(p)$ is *independent of the decoder* is a consequence of our framework—$\Delta D(s)$ depends only on the noise channel $\mathcal{N}$ and the code state $\sigma_{\mathcal{C}}$, not on the recovery strategy. Recent evidence supports this structure: Boltzmann-factor failure rates via the Ising–QEC mapping [21], statistical-mechanical thresholds [20], and scalable post-selection gains [19, 22]. See [31], Sec. S9 for derivation.

Prediction 2: Decoder hierarchy as retrodiction approximation.—Every decoder $\mathcal{D}$ has a retrodiction gap $\delta_{\mathcal{D}} = D(\tilde{\mathcal{R}}_{\sigma,\mathcal{N}} \| \mathcal{D})$. *What is new:* if the Petz map were merely one good recovery map among many, $\delta_{\mathcal{D}}$ would have no reason to correlate with decoder performance. But the uniqueness theorem (Theorem 1) implies that *all* good decoders must approximate the Petz map, so $\delta_{\mathcal{D}}$ must rank-order decoder performance. The known decoder hierarchy [23]—ML > neural network [24] > MWPM > Union-Find—should therefore be a hierarchy of retrodiction fidelity. See [31], Secs. S10 and S12 for the derivation and numerical verification.

THE QUANTUM ERASER REVISITED

The quantum eraser is a pure bipartite system: $\mathcal{N} = \mathrm{Tr}_E$, $\sigma = |\psi\rangle\langle\psi|_{SE}$. As computed in [31] (Sec. S1), the Petz map achieves $F = 1$, hence $\tau = 0$. This is a direct consequence of the equivalence chain (11): for a pure bipartite state, $I(A; E|B) = 0$ (no environment beyond the idler) and $\Sigma = 0$ (no dissipation). The quantum eraser "works" because the signal-idler pair forms a closed system with $\tau = 0$.

Superposition necessity.—By contrapositive: if a system is in a definite path state, the Petz recovery produces no coherence (see [31], Sec. S5). Recovery of interference implies prior superposition.

Wheeler's delayed choice and τ.—In Wheeler's delayed-choice experiment [15], the signal-idler pair forms a time-symmetric ($\tau = 0$) system where forward and reverse processes are statistically indistinguishable. The *order* of measurements is operationally irrelevant: measuring the signal first or the idler first yields identical statistics. The delayed-choice aspect is demystified—not by invoking retrocausality, but by recognizing that the closed system has no arrow of time. The "need to wait" for the idler measurement arises only in the open-system description, where one traces out the idler and observes $\tau > 0$ (decoherence) [27].

DISCUSSION

We have shown that the Petz recovery map $\mathcal{R}_{\sigma,\mathcal{N}}$ simultaneously describes quantum erasure, time reversal, error correction, and thermodynamic irreversibility. The temporal asymmetry parameter $\tau = 1 - F$ (Eq. (9)) provides a physical interpretation: it measures the degree to which a quantum process possesses an arrow of time.

Domain	**Key bound**	τ
Quantum eraser	$F = 1$ (exact)	0
Closed unitary	Uniqueness	0
QEC	$F^2 \geq e^{-\Delta D}$	> 0
Thermodynamics	$F^2 \geq e^{-\beta W_{\mathrm{diss}}}$	$\leq 1 - e^{-\Sigma/2}$

The closed-system rows have $\tau = 0$: the Petz map is exact, recovery is perfect, and there is no physical distinction between "before" and "after" the channel. The open-system rows have $\tau > 0$, bounded by entropy production via Eq. (10). This connects to the decoherence program [27, 28]: environment-induced decoherence produces entropy ($\Sigma > 0$), which implies $\tau > 0$, which breaks time-reversal symmetry. The arrow of time is not a postulate but a consequence of system-environment coupling, quantified by the Petz recovery fidelity.

Moreover, Eq. (10) yields a *retrodiction Landauer principle* (see [31], Sec. S11): the minimum energy cost of retrodicting the past from the future is $E_{\mathrm{retro}} \geq k_{\mathrm{B}} T \cdot \Sigma$. Retrodiction is thermodynamically free if and only if the process is reversible.

Open questions.—(i) *Approximate uniqueness*: If axiom (3) holds only approximately, is the resulting map close to the Petz map? (ii) *Holographic implications*: Since entanglement wedge reconstruction uses the Petz map [8], Theorem 3 implies a thermodynamic bound on bulk reconstruction fidelity. (iii) *Post-selection experiments*: The prediction $\ln R(\varepsilon, d) = \alpha d + \beta(\varepsilon)$ is directly testable on current hardware [25]. (iv) *Decoherence and*

τ: Can τ serve as a quantitative measure of the quantum-to-classical transition [27]? (v) *Non-Markovian dynamics*: For non-Markovian processes, Σ can be temporarily negative [30], implying transient decreases in τ—partial "time reversal." The process tensor framework [16] may extend the equivalence chain (11) to this regime.

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[31] See Supplemental Material for detailed proofs of all theorems, the explicit quantum eraser calculation, post-selection analysis, the decoder hierarchy, numerical verification of Predictions 1 and 2, the equivalence chain proof, and caveats.